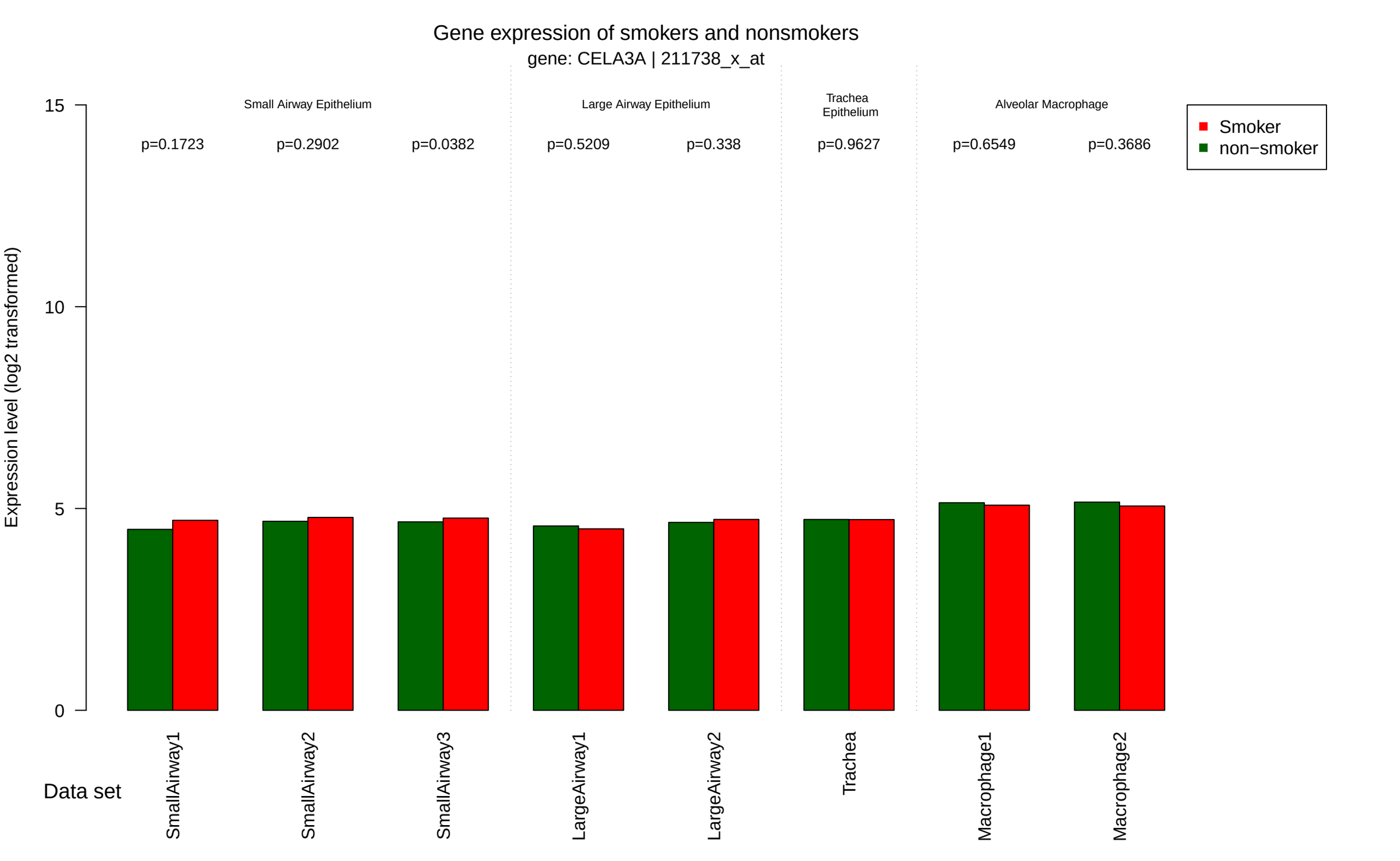
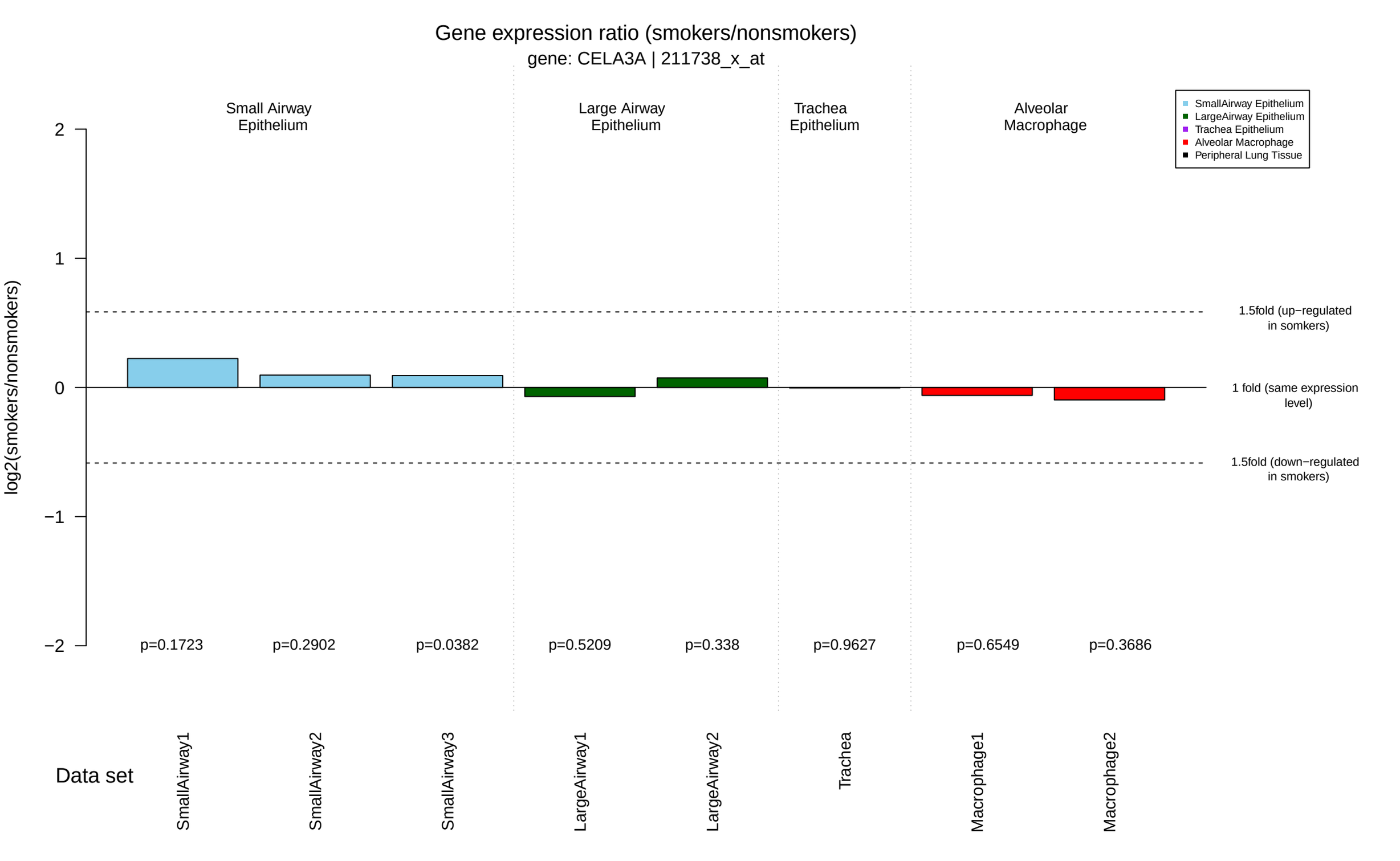
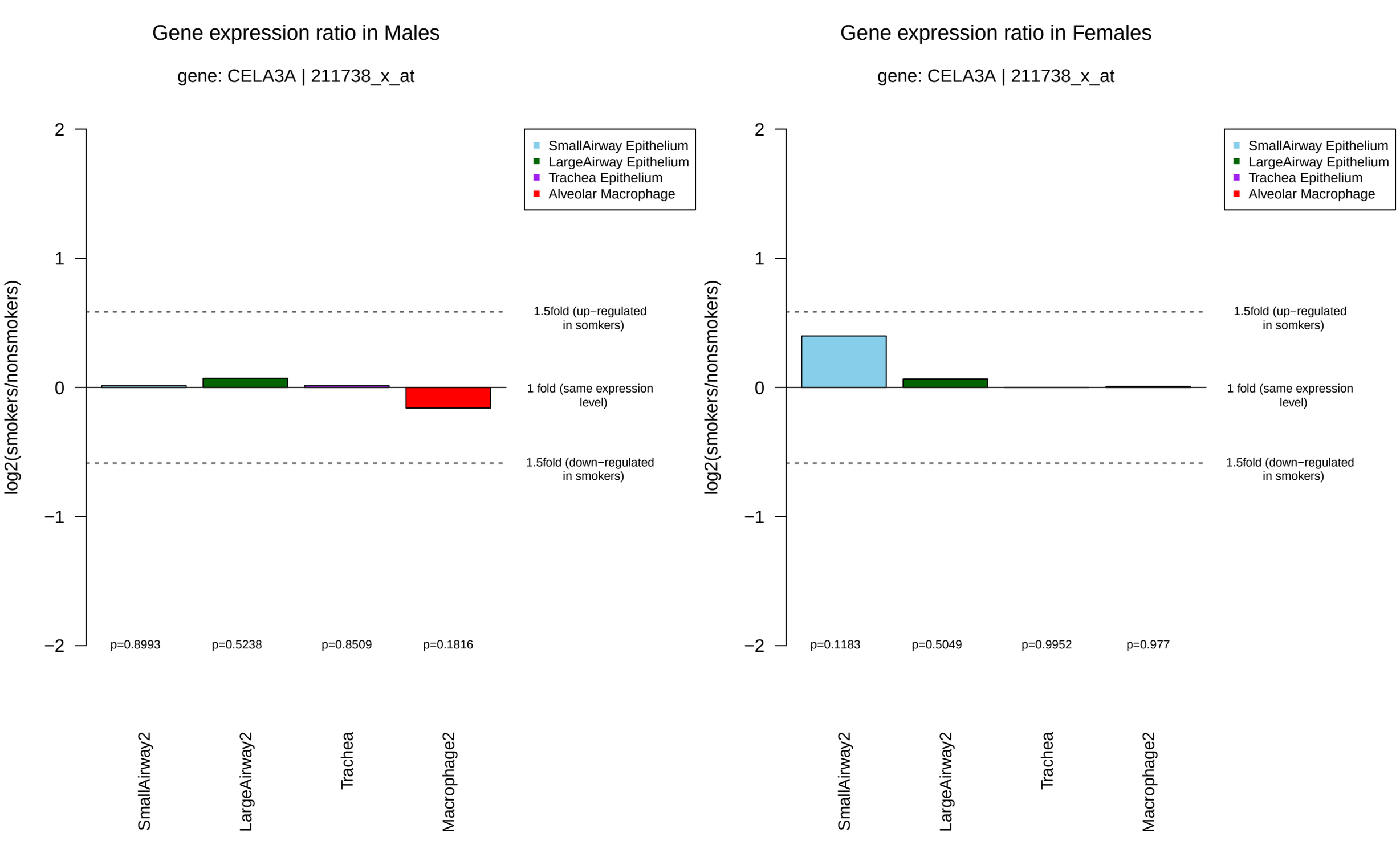
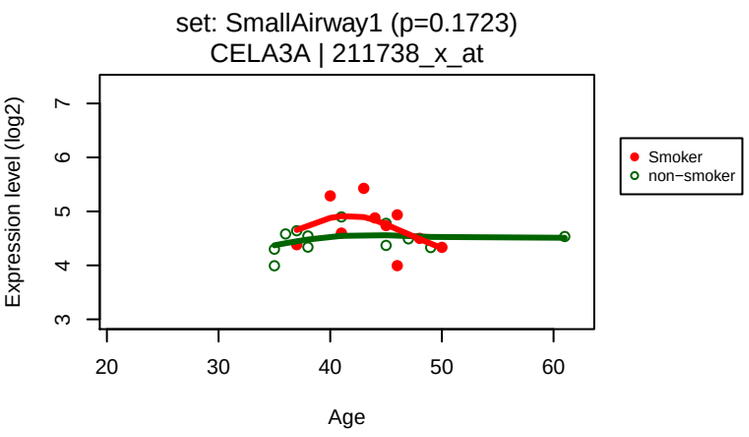
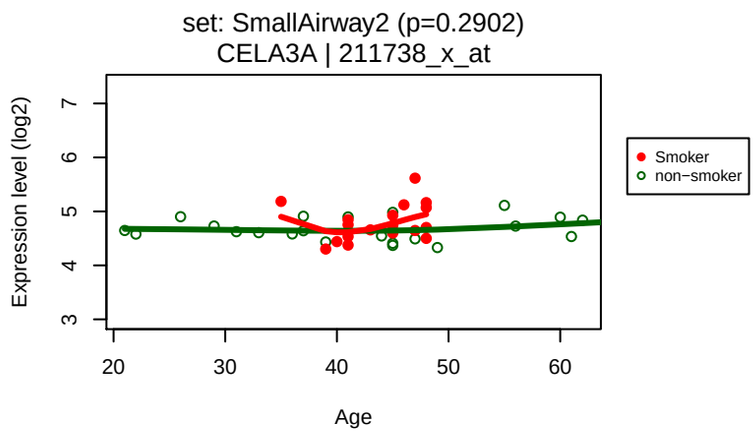
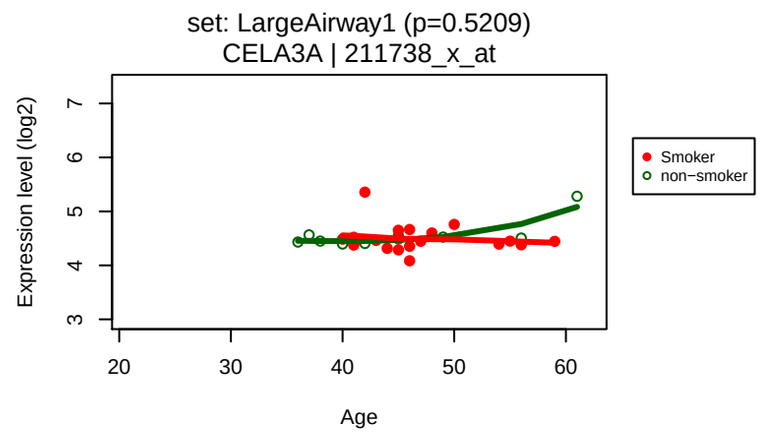

Gene expression of smokers and nonsmokers
gene: CELA3A | 211738_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1723
p=0.2902
p=0.0382
p=0.5209
p=0.338
p=0.9627
p=0.6549
p=0.3686
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

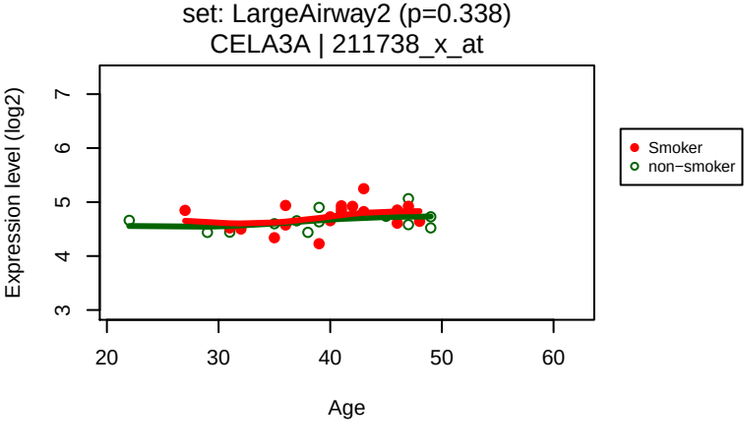

Gene expression ratio (smokers/nonsmokers)
gene: CELA3A | 211738_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1723
p=0.2902
p=0.0382
p=0.5209
p=0.338
p=0.9627
p=0.6549
p=0.3686
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

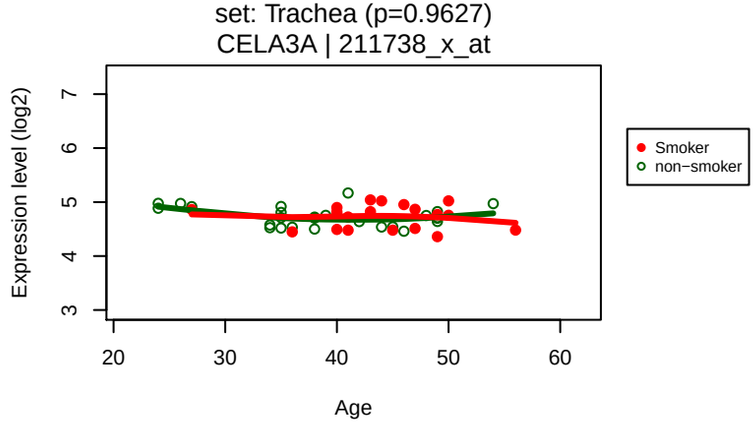

Gene expression ratio in Males
gene: CELA3A | 211738_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8993
p=0.5238
p=0.8509
p=0.1816
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CELA3A | 211738_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1183
p=0.5049
p=0.9952
p=0.977
SmallAirway2
LargeAirway2
Trachea
Macrophage2

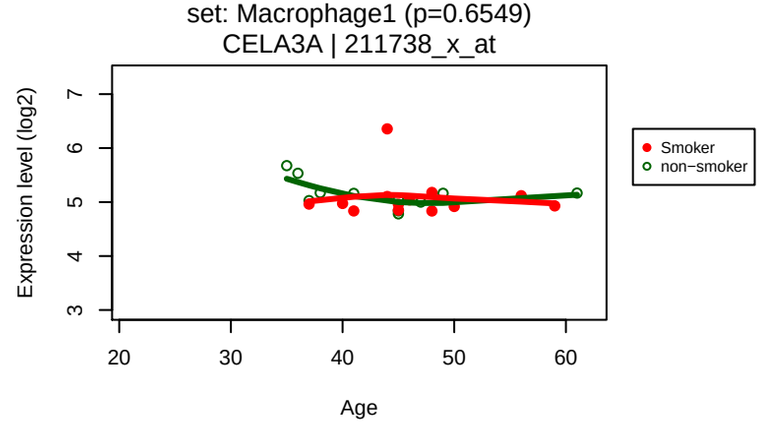

set: SmallAirway1 (p=0.1723)
CELA3A | 211738_x_at
Expression level (log2)
Age
Smoker
non−smoker

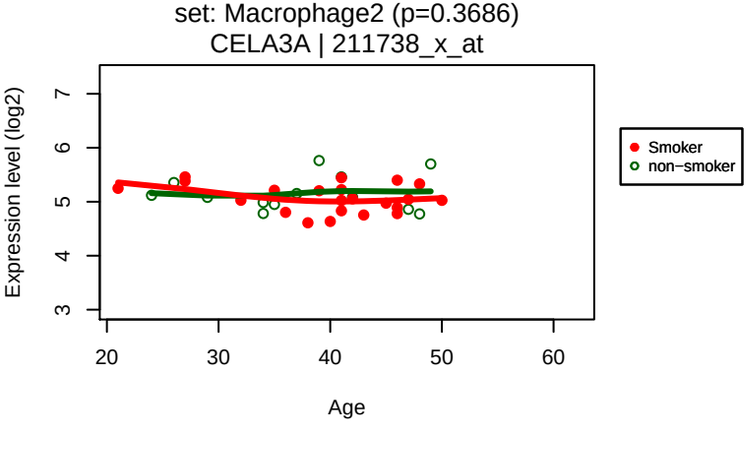

set: SmallAirway2 (p=0.2902)
CELA3A | 211738_x_at
Expression level (log2)
Age
Smoker
non−smoker

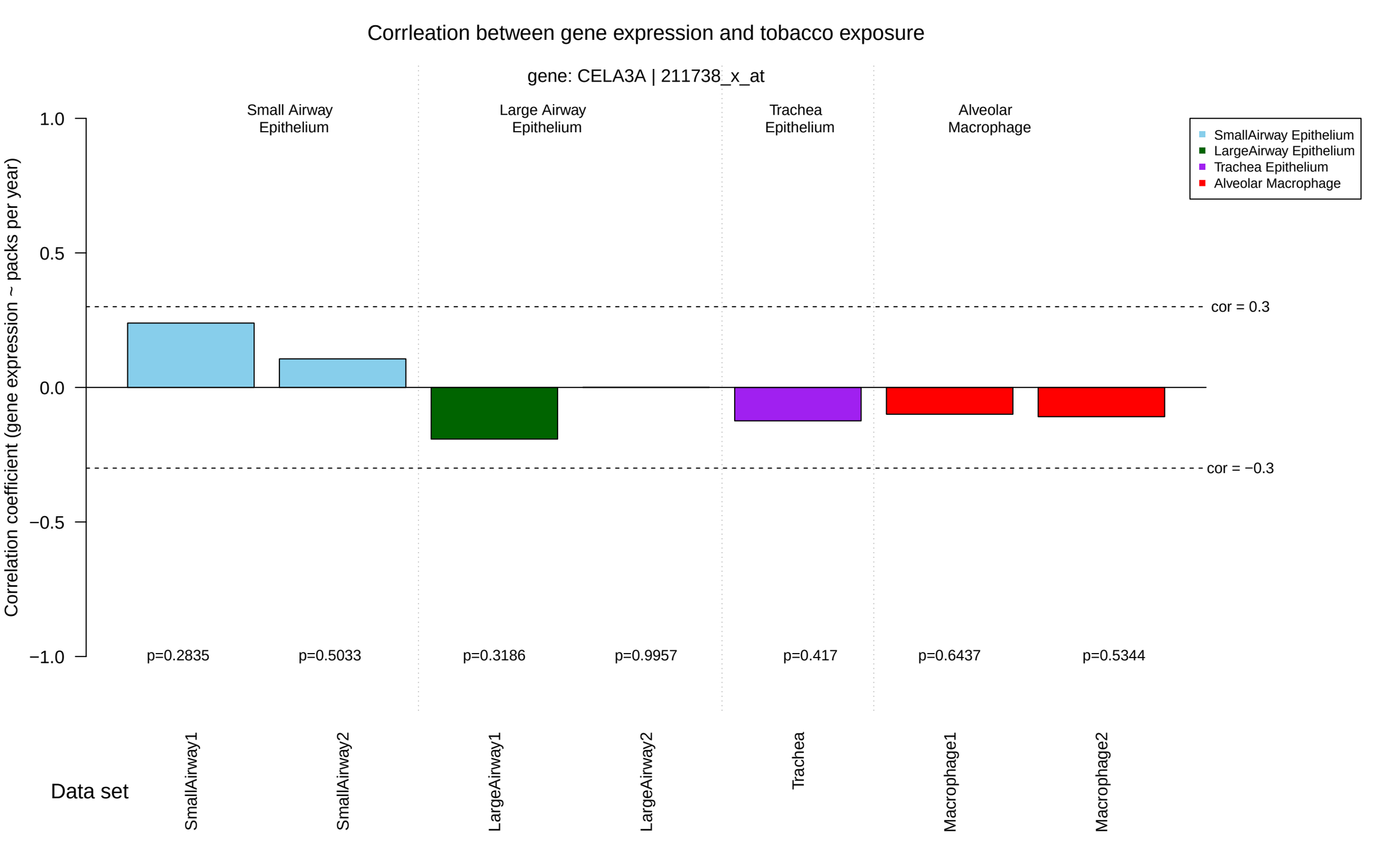

set: LargeAirway1 (p=0.5209)
CELA3A | 211738_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.338)
CELA3A | 211738_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9627)
CELA3A | 211738_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6549)
CELA3A | 211738_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3686)
CELA3A | 211738_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CELA3A | 211738_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2835
p=0.5033
p=0.3186
p=0.9957
p=0.417
p=0.6437
p=0.5344
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2